

UCE799: PROJECT SEMESTER				
	L	T	P	Cr
	-	-	-	15.0
Course Objective: The objective of the six month industrial training is to expose the final year civil engineering students to the competency, knowledge and skills needed to succeed at the workplace. By undergoing industrial training, they will be able to relate the theory that they learnt and applied them practically. Industrial Training is essential for students to develop the practical skills that they will need to be effective professional engineers				
Course Introduction: The project semester is aimed at developing the undergraduate education programme in engineering to include a practical training in a professional engineering setting (a company, top educational institution, research institute etc.) hereafter referred to as host “organization” as deemed appropriate. The participating organizations are selected that are either already visiting Thapar University for placement or are forming new relationships of mutual benefit. The project semester gives the student the opportunity to translate engineering theory into practice in a professional engineering environment. A central requirement of the project semester is that it must be based around significant engineering work and is principally assessed on that basis. The technical activity should be related to both the student’s engineering studies and to the host organization’s activities, and it should constitute a significant body of engineering work at the appropriate level. It should involve tasks and methods that are more appropriately completed in a professional engineering environment and should, where possible, make use of human and technology resources provided by the organization. It consolidates the student’s prior learning and provides a context for later research studies. The student remains a full time registered student at ThaparInstitute during the project semester and this activity is therefore wholly distinct from any industrial interactions which may occur over vacation periods				
Course Learning Objectives (CLO) The project work undertaken as part of the project semester is diverse. As a result, the Learning Outcomes will vary, but on completion of the module, students will have achieved several learning outcomes from the following list: 1. Able to identify and use appropriate mathematical methods, numerical techniques and software tools for application to new and ill-defined engineering problems; 2. Be able to integrate knowledge, handle complexity and formulate judgements with incomplete or limited information; 3. Have the ability to redesign products, processes or systems in order to improve productivity, quality, safety and other desired needs; 4. Have the ability to apply design methods, processes and techniques to unfamiliar, ill-defined problems, involving other disciplines; 5. Be able to design according to codes of practice and industry standards; to identify limitations of codes of practice and the need for their application 6. Have the ability to investigate and define a need and identify constraints including health, safety and legal issues and the impact of engineering solutions in a societal and environmental				

context;

7. Be able to make engineering judgements that take cognizance of the social, environmental, ethical, economic, financial, institutional and commercial considerations affecting the exercise of their engineering discipline;

8. Have the ability to consult and work with experts in various fields in the realization of a product or system;

9. Have knowledge and understanding of concepts from a range of areas outside engineering;

10. Be able, via knowledge and understanding of group dynamics, to exercise leadership;

11. Be able to select and apply appropriate communication tools and write technical papers and reports;

12. Be able to describe the relevant advantages and disadvantages of various technologies to an audience, and to communicate effectively in public.

Evaluation Scheme:

Each student is assigned a faculty supervisor who is responsible for managing and assessment of the project semester. This includes a Reflective Diary which is updated throughout the project semester, an Interim Project Report, a Final Report with Learning Agreement/Outcomes and a Final Presentation & Viva which involves the faculty Supervisor and some other members from the department. A hard copy and electronic copy of all reports are required. The mentor from the host organization will be asked to provide his assessment on the designated form. A suggested weighting for the assessments is as follows:

Activity	Submission time line	Marks awarded by	Weightage (%)
Reflective Diary	End of Project Semester	Faculty Supervisor	10
Goals Report	End of week 4 of project semester	Faculty Supervisor	5
Midway Report	End of week 10 of project semester	Faculty Supervisor	15
Final Assessment	End of Project Semester	Host Mentor	20
Final Report	End of Project Semester	Committee assessment	20
Oral and Poster presentation and viva	End of Project Semester		30